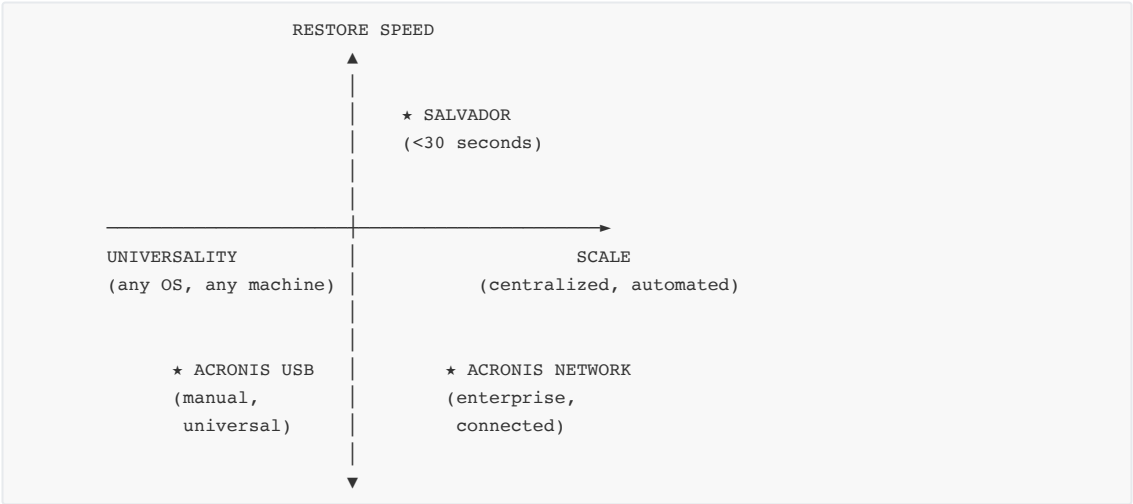


Solution Comparison: Strategic View

Supporting material for Cyber Team discussion

The Core Tradeoff

Every backup solution makes a fundamental tradeoff. You can optimize for speed, scale, or universality, but not all three. Our framework accepts this reality and assigns each tool to where it excels.



The strategic insight: We're not choosing the "best" solution. We're building a portfolio where each tool handles what it does best.

Salvador: The Surgical Instrument

When to Deploy

Salvador exists for one scenario: machines where downtime cost exceeds backup cost, and where speed of recovery is the primary constraint.

A furnace control PC costs €50,000 per day when down. Salvador costs €1,000 per year. The math is obvious. What's less obvious is that Salvador only makes sense when ALL of these are true:

- Restore time matters (production-critical)
- Agent can be installed (Windows, physical access)
- Network is not available OR air-gap is preferred
- Cost is justified by criticality

If any of these conditions fail, another solution is better.

The Security Proposition

Salvador's security model is isolation. The backup device sits inside the machine, disconnected from any network. There is no attack surface from the network. The only threat vector is physical access to the machine itself.

This is appealing for high-value targets. A ransomware attack cannot encrypt a Salvador backup. A network intrusion cannot reach it. The tradeoff is that you also cannot manage it remotely without the optional network monitoring feature.

Question for cyber team: Is physical isolation a genuine security advantage in our environment, or does it create blind spots in our visibility?

The Operational Reality

Salvador requires per-machine installation. There is no central dashboard without network connectivity. Each machine is an island. For a small number of critical machines, this is manageable. For hundreds, it becomes operationally hard to manage.

This is why Salvador is a scalpel, not a sledgehammer. We estimate 80-100 machines at Pomigliano where the tradeoff makes sense.

Acronis Network: The Workhorse

When to Deploy

Acronis Network is the default choice for any machine that is:

- Network-connected (or worth connecting)
- Running Windows or Linux
- Able to accept an agent installation
- Not so critical that 3-hour restore time is unacceptable

This covers roughly 60% of the shop floor fleet.

The Security Proposition

Acronis Network moves data over the internal network to a central server. This creates both advantages and risks.

Advantages:

- Centralized visibility into backup status
- Configurable retention policies
- Compliance reporting
- Single point of management

Risks:

- Data in transit over network (requires encryption)
- Central server is a high-value target
- Agent installation expands attack surface on endpoints
- Network segmentation becomes critical

Question for cyber team: We saw data isolation challenges in our trials. What's the minimum network architecture to make this acceptable?

The Operational Reality

Setting up Acronis Network took eight months. Not because the product is complex, but because our environment is. Firewall rules, server provisioning, security approvals, network segmentation discussions. Once operational, it runs reliably with minimal intervention.

The lesson: deployment cost is front-loaded. After setup, marginal cost per machine is low.

Acronis USB: The Universal Fallback

When to Deploy

Acronis USB exists because some machines cannot run modern software. A Fanuc CNC controller from 2005 doesn't care about your enterprise backup strategy. It runs a proprietary OS and it doesn't have a network port.

USB backup is the answer when:

- OS is legacy or proprietary (DOS, Windows 95/98, Fanuc, Makino, etc.)
- Agent installation is impossible
- Network connectivity doesn't exist
- The machine is low enough criticality that manual process is acceptable

This is not a preferred solution. It's a necessary one.

The Security Proposition

USB backup is air-gapped by nature. Data never touches a network. The backup exists on physical media that must be stored securely.

Advantages:

- No network exposure
- Works with any OS that can boot from USB
- Simple, understandable process

Risks:

- Physical media can be lost or stolen
- No centralized tracking of backup status
- Human error in the backup process
- Backup freshness depends on maintenance discipline

Question for cyber team: For legacy machines that can't run agents, is USB acceptable? Are there specific handling requirements for the physical media?

The Operational Reality

USB backup is labor-intensive. Someone must physically visit the machine, connect the drive, run the backup, store the drive securely. For 100+ machines, this is a significant operational burden.

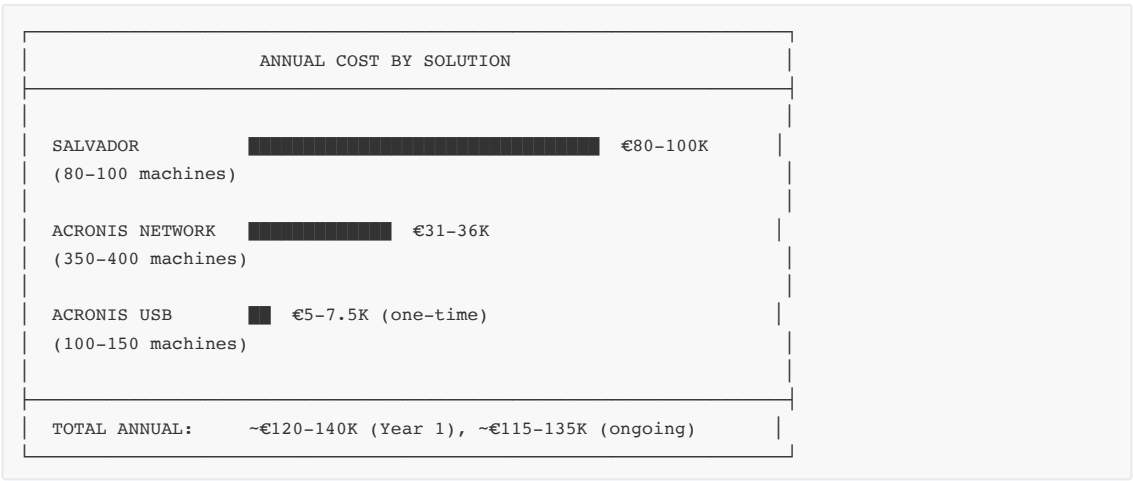
This is why USB is the last resort, not the first choice. We estimate 100-150 machines at Pomigliano where no other option exists.

Comparative Summary

Dimension	Salvador	Acronis Network	Acronis USB
Primary Value	Speed	Scale	Universality
Restore Time	30 seconds	3 hours	3 hours
Annual Cost	€1,000/machine	€90/machine	€50 one-time
Management Model	Per-machine	Centralized	Manual
OS Coverage	Windows	Win + Linux	Universal
Network Dependency	None	Required	None
Security Model	Physical isolation	Network + encryption	Physical media
Scalability	Low (manual install)	High (enterprise)	Very low (manual)
Our Target Deployment	80-100 machines	350-400 machines	100-150 machines

The Cost Picture

Investment is not about minimizing spend. It's about right-sizing investment to risk.



The strategic argument: €120K/year protects €600M+ in annual production value. A single major incident (ransomware, hardware failure without backup) could cost more than a decade of backup investment.